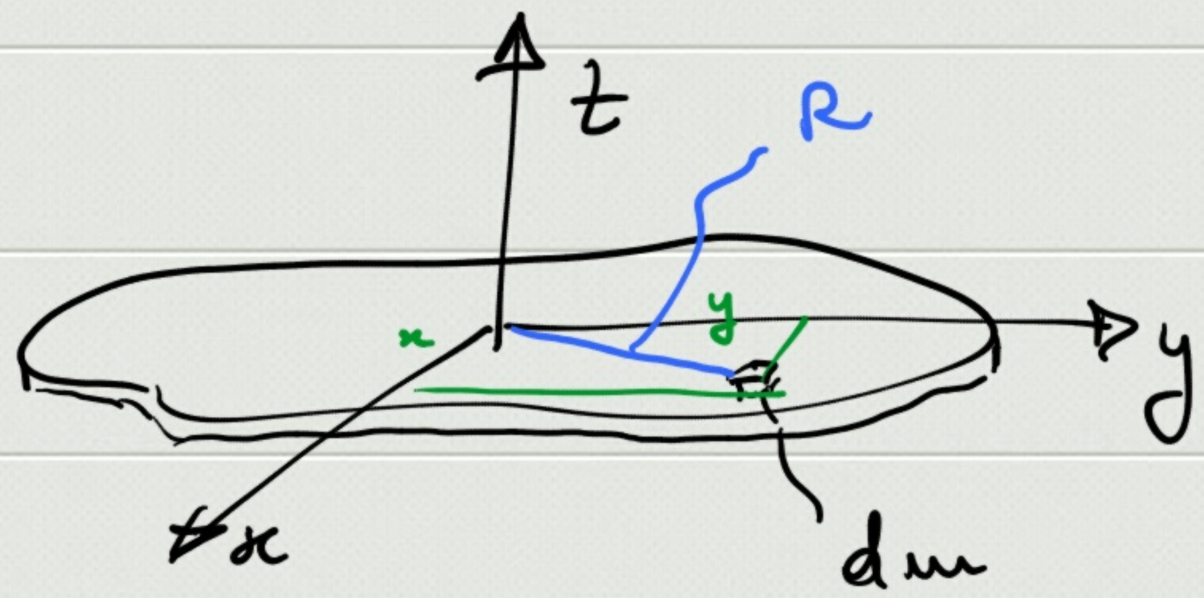


Figure piana

$$I = \int R^2 \, dm$$

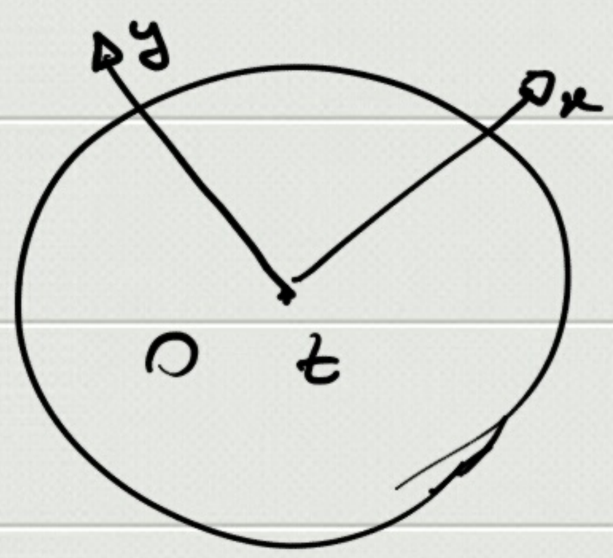
$$I_z = \int (x^2 + y^2) \, dm$$

$$I_x = \int R_x^2 \, dm = \int (y^2 + \underset{0}{\underset{\shortparallel}{z^2}}) \, dm = \int y^2 \, dm$$

$$I_y = \int R_y^2 \, dm = \int (x^2 + \underset{0}{\underset{\shortparallel}{z^2}}) \, dm = \int x^2 \, dm$$

$$\boxed{I_z = I_x + I_y}$$

Teorema delle figure piane



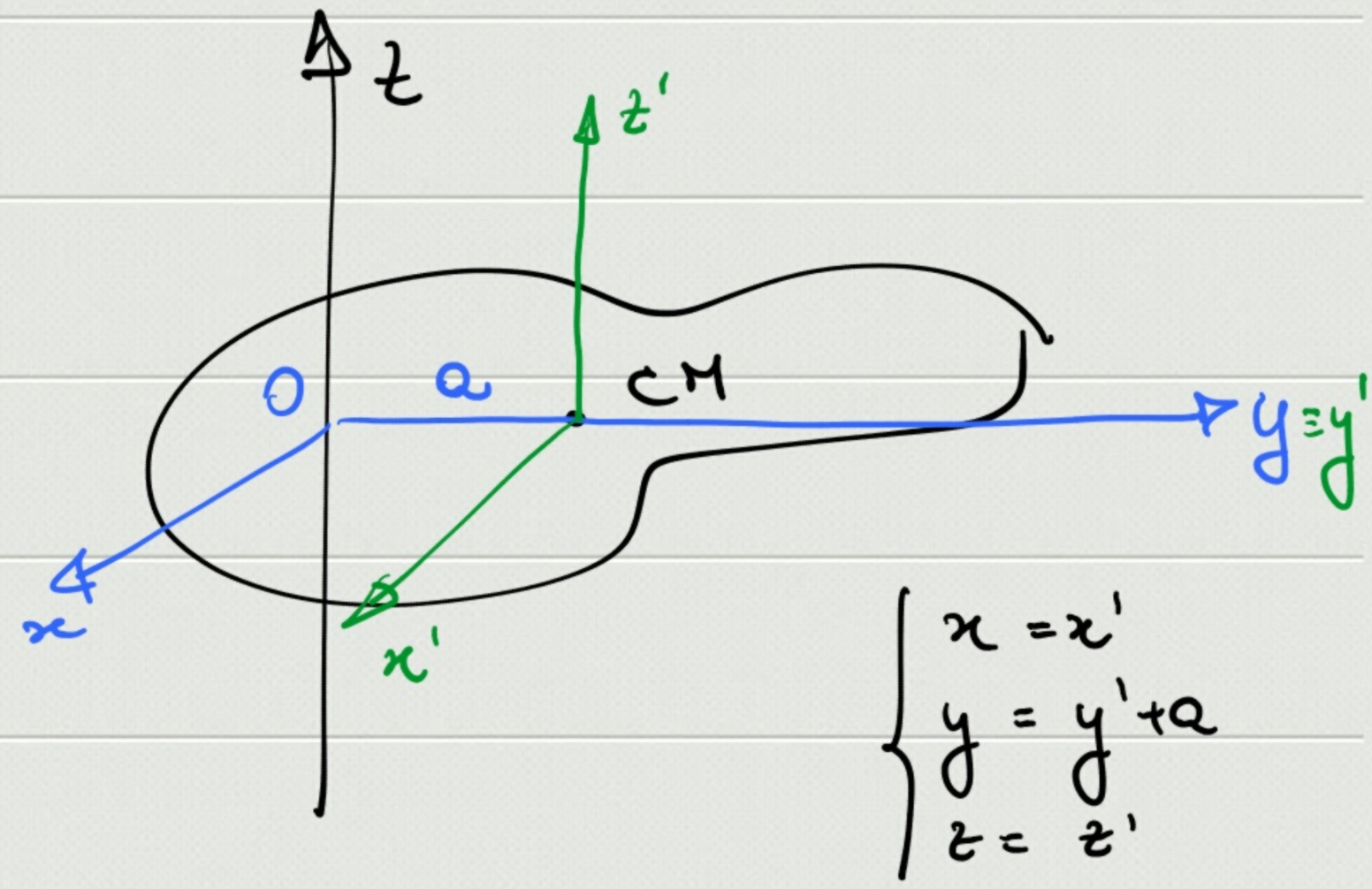
Anello(m, R) $\quad I_z = mR^2$

$$I_x = I_y$$

$$\Rightarrow I_z = I_x + I_y = 2 I_x$$

$$\Rightarrow I_x = \frac{1}{2} I_z = \frac{1}{2} m R^2$$

$I_z = ?$



$$\begin{cases} x = x' \\ y = y' + a \\ z = z' \end{cases}$$

$$I_z = \sum_i m_i R_i^2 = \sum_i m_i \left(x_i^2 + y_i^2\right) =$$

$$= \sum_i m_i \left[x_i'^2 + (y_i' + a)^2\right] =$$

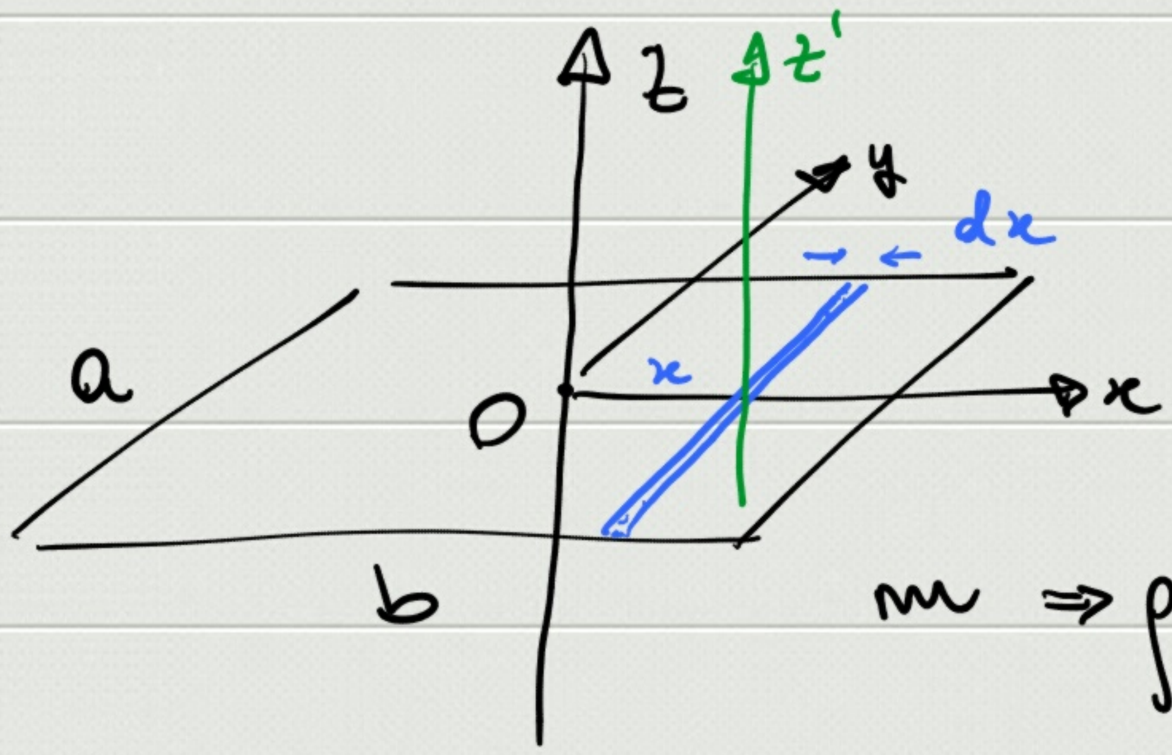
$$= \sum_i m_i \left(x_i'^2 + y_i'^2 + a^2 + 2a y_i'\right) =$$

$$= \sum_i m_i \underbrace{\left(x_i'^2 + y_i'^2\right)}_{R_i'^2} + \sum_i m_i a^2 + \cancel{2a \underbrace{\sum_i m_i y_i'}_{y'_{CM} \cdot m \,\hookrightarrow\, = 0}} =$$

$$= I_{z'} + m a^2$$

$$\boxed{I_z = I_{z'} + m a^2}$$

Teorema di Huygens Steiner



$$I_z = \int R^2 \, dm$$

$$m \Rightarrow \rho_S = \frac{m}{a\,b}$$

$$I_z = \int dI_z \qquad \left[I_{sbarra} = \frac{1}{12} m d^2 \right]$$

$$I_{z,lastra} = \int dI_{z,sbarra} \overset{H\text{-}S}{=} \int \left(dI_{z',sbarra} + x^2 dm \right) =$$

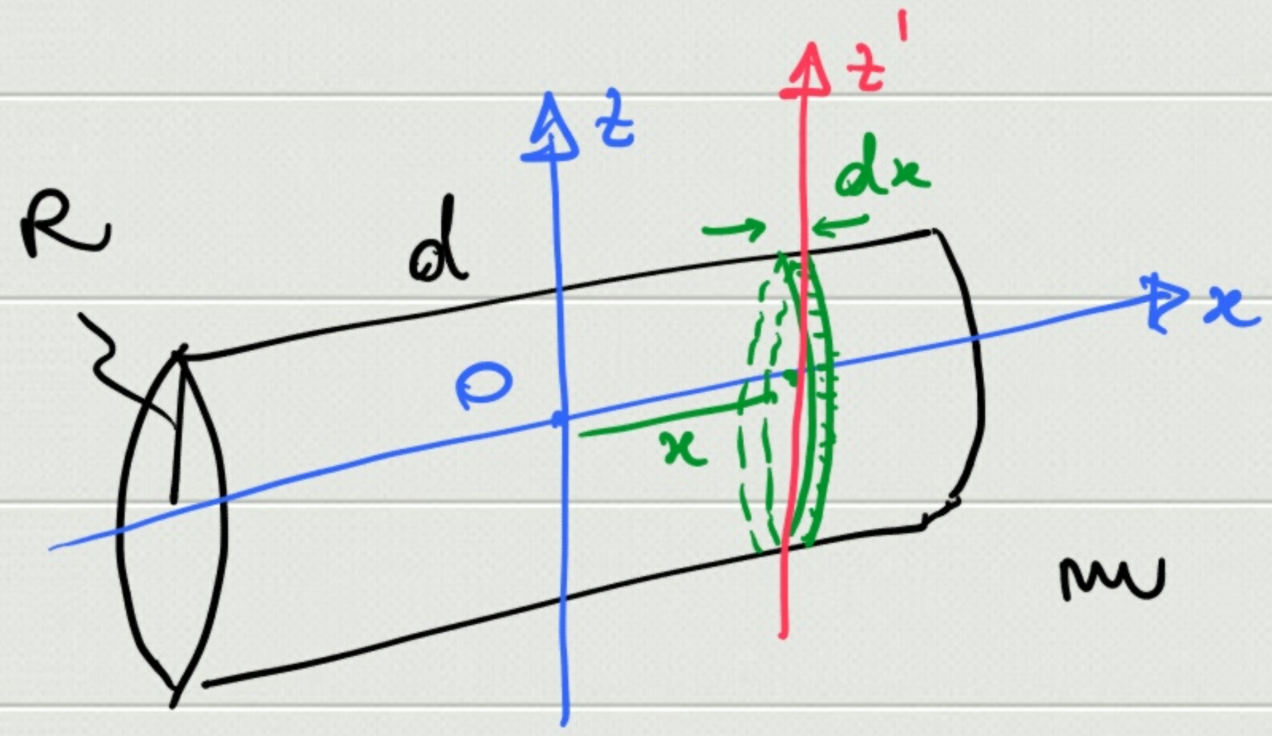
$$= \int \left(\frac{1}{12} dm\, a^2 + x^2 \rho_S \, dS \right) =$$

$$= \int \frac{1}{12} a^2 dm + \int_{-b/2}^{b/2} x^2 \frac{m}{\cancel{a} b} \cancel{a} \, dx =$$

$$= \frac{1}{12} m a^2 + \frac{m}{b} \left[\frac{x^3}{3} \right]_{-b/2}^{b/2} =$$

$$= \frac{1}{12} m a^2 + \frac{m}{3b} \left(\frac{b^3}{8} + \frac{b^3}{8} \right) =$$

$$= \frac{1}{12} m a^2 + \frac{m}{3\cancel{b}} \frac{b^{\cancel{3}\,2}}{4} = \underline{\underline{\frac{1}{12} m \left(a^2 + b^2 \right)}}$$



guscio sottile

$$I_z = \int dI_z \underset{\text{anello}}{\overset{H\text{-}S}{=}} \int \left(dI_{z'} + x^2 dm \right) =$$

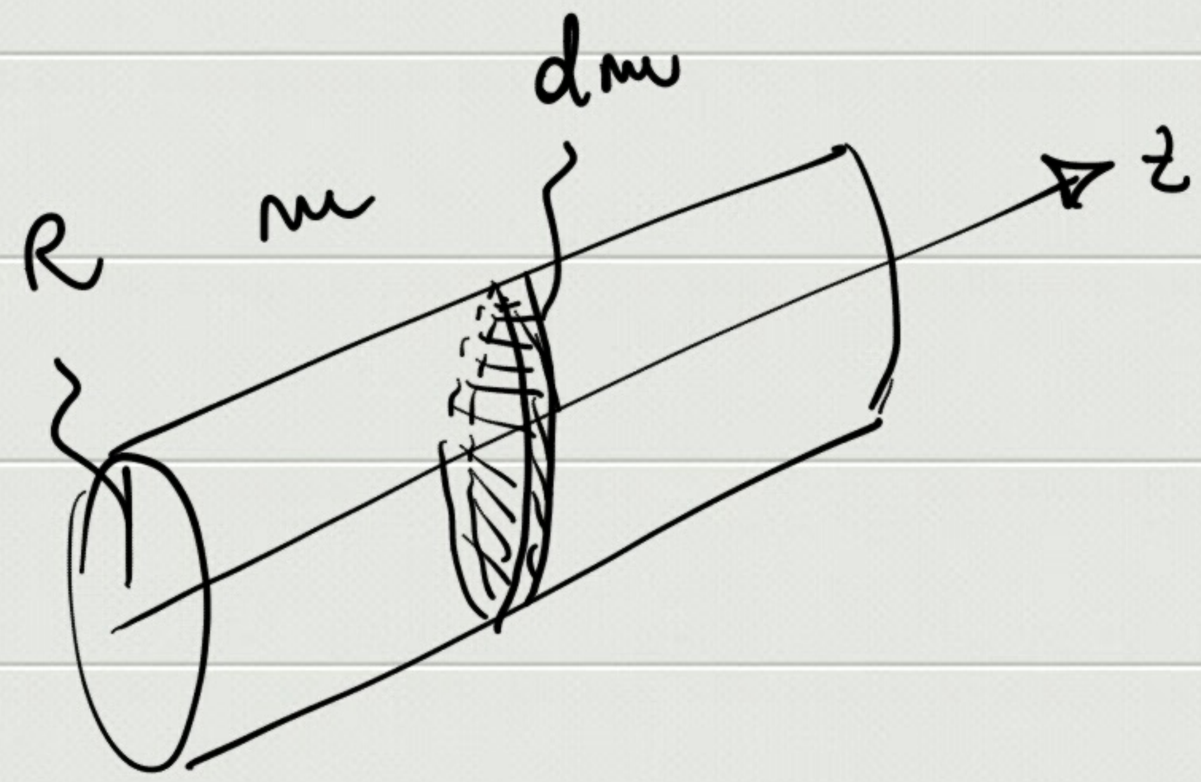
$$= \int \left(\frac{1}{2} dm R^2 + x^2 dm \right) = \frac{1}{2} R^2 m + \int x^2 dm =$$

$$dm = \rho_S \, dS = \frac{m}{\cancel{2\pi R} d} \cancel{2\pi R} \, dx = \frac{m}{d} dx$$

$$= \frac{1}{2} m R^2 + \int_{-d/2}^{+d/2} x^2 \frac{m}{d} dx =$$

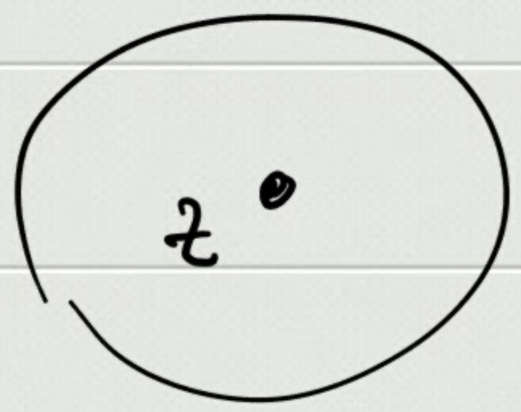
$$= \frac{1}{2} m R^2 + \frac{m}{d} \left[\frac{x^3}{3} \right]_{d/2}^{d/2} =$$

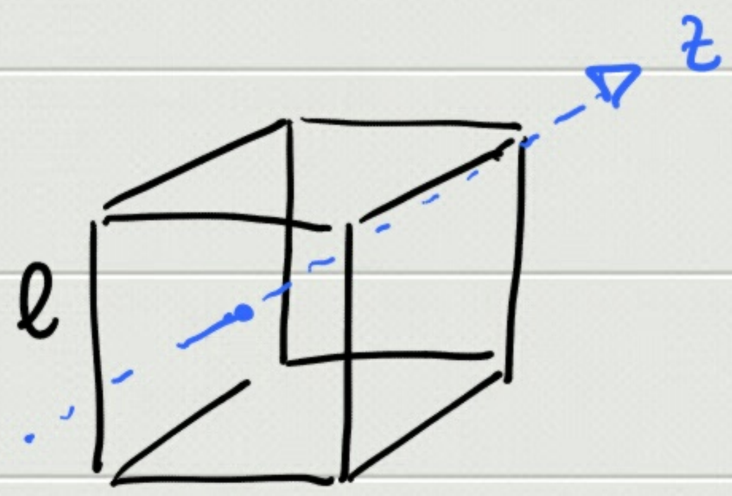
$$= \frac{1}{2} m R^2 + \frac{m}{3\cancel{d}} \frac{d^{\cancel{3}\,2}}{4} = \frac{1}{2} m R^2 + \frac{1}{12} m d^2$$

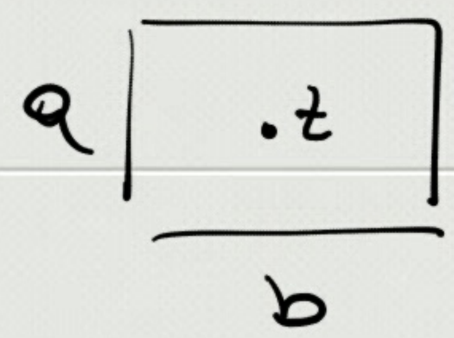


cilindro pieno

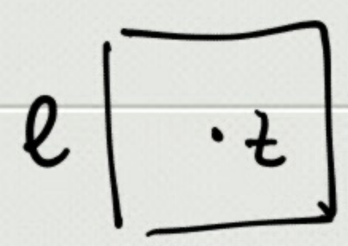
$$\underset{\text{cilindro}}{I_z} = \int \underset{\substack{\downarrow \\ \text{disco}}}{dI_z} = \int \frac{1}{2} dm R^2 = \frac{1}{2} R^2 \int dm = \underline{\underline{\frac{1}{2} m R^2}}$$







$$I_z = \frac{1}{12} m (a^2 + b^2)$$



$$I_z = \frac{1}{12} m (l^2 + l^2) = \frac{1}{6} m l^2$$

$$\boxed{I_{z,\text{cubo}} = \frac{1}{6} m l^2}$$